

Program : Diploma in Computer Hardware Engineering	
Course Code : 6152	Course Title: Computer System Hardware
Semester : 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- This course is designed to get a thorough knowledge of internal and external components of desktop computers.
- Students will be able to describe the functioning of a computer system.
- To Demonstrate hardware components, setup/ configuration of laptop computers.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics on Computer Fundamentals		Introduction to IT systems lab.	1

Course Outcomes :

On completion of the course student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate basic structure of a computer, Motherboard components of desktop computers and BIOS settings.	14	Understanding
CO2	Demonstrate power supply and storage devices of a computer	15	Understanding
CO3	Demonstrate working of input-output devices.	15	Understanding
CO4	Demonstrate motherboard components, hardware setup/configuration of peripherals in laptop computers	14	Understanding
	Series Tests	2	

CO – PO Mapping

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate basic structure of a computer, motherboard components in desktop computers and BIOS settings		
M1.01	Illustrate the functional units of a computer with block diagram	3	Understanding
M1.02	Describe different connectors available in desktop computers	3	Understanding
M1.03	Demonstrate motherboard components and its features in desktop computer	4	Understanding
M1.04	Demonstrate BIOS settings	4	Understanding
Contents: Functional units of a computer- Block diagram Desktop motherboard components : Motherboard form factors, Chipset, CPU Socket types, CPU fan & Heatsink mounting points, Connectors for integrated peripherals, CMOS Backup battery, Bus & Interconnection slots, South bridge, North bridge, AGP slots, IDE & SATA Connectors- features, Memory slots, Switches & Jumpers. Overview of SMPS and its features. BIOS : Define booting & understand the concept of multi os booting, BIOS – settings & configuration			
CO2	Demonstrate power supply and storage devices of a computer		
M2.01	Describe characteristics of Power Supplies.	2	Understanding
M2.02	Explain the details about powering the PC	2	Understanding
M2.03	List memory hierarchy	2	Understanding
M2.04	Demonstrate different types of primary memory	4	Understanding

M2.05	Demonstrate different types of secondary memory devices	5	Understanding
	Series Test – I	1	
Contents: Purpose and characteristics of Power Supplies- Power connectors- Replacement of Power Supplies Introduction to computer memory system, Memory characteristics & Hierarchy. Different types of memory – Primary memory : Semiconductor memory technology, RAM, ROM, Cache memory concepts & levels-Secondary memory – Direct access, Random access, Sequential access storage devices.			
CO3	Demonstrate the working of input- output devices		
M3.01	Describe different categories of input and output devices.	3	Understanding
M3.02	Describe characteristics and working of input devices	6	Understanding
M3.03	Describe characteristics and working of output devices	6	Understanding
Contents: List & Explain different categories of input & output devices : Physical connection, Virtual connection, Shared, and Dedicated. Input devices: Keyboard, Mouse, Joystick, Lighten, Microphone, Scanner, Bar code Reader etc.-Output devices : Monitor, Printer, Speaker, Projector, Headset etc			
CO4	Demonstrate hardware components, setups /configurations of peripherals in laptop computers		
M4.01	Demonstrate hardware components of laptop computers and its features	4	Understanding
M4.02	Demonstrate BIOS Setup and configuration operations in laptop systems.	4	Understanding
M4.03	Demonstrate disassemble & assemble laptop with parts	3	Understanding
M4.04	Demonstrate Laptop Repairing methods.	3	Understanding
	Series Test – II	1	
Contents: Laptop motherboard components: Top panel, Base panel, Touch pad, Palm rest assembly, Hinges, CPU, Processors, Primary Memory and Secondary Storage, Screen, Graphics and Sound Card , Batteries, Adapters, TFT/LCD/LED, Inverter, CCFL, wifi card-BIOS SETUP, TPM Configuration, CPU Configuration, SATA Configuration, USB Configuration, On Board Device Configuration, Power management configuration. Disassembling and Assembling a laptop with components-Repairing Concepts: Adapter repairing, Battery repairing, Keyboard/ Touchpad, Hard Disk, RAM etc			

Text / Reference

T/R	Book Title/Author
T1	Stephen J. Bigelow, Troubleshooting, Maintaining and Repairing PCs, TMH, Fifth Edition.
T2	Morris Rosenthal, An introduction to Troubleshooting and Repairing laptop.
T3	D Balasubramanian., Computer installation and servicing. TMH, 2010
R1	Complete A+ Guide to IT Hardware and Software: AA CompTIA A+ Core 1 (220-1001) & CompTIA A+ Core 2 (220-1002)
R2	Joel Rosenthal , PC Repair and Maintenance- Fire wall Media, First Edition
R3	Manahar Lotia, Pradeep Niar, Modern Computer Hardware Course BPB Publication ,2011
R4	Robert Bruce and Co O' Reilly ,PC Hardware in a nutshell, Shroff Publishers and Distributors -2008

Online Resources

S.No	Website Link
1	https://nptel.ac.in/courses/
2	https://www.khanacademy.org
3	https://ocw.mit.edu/courses